



EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



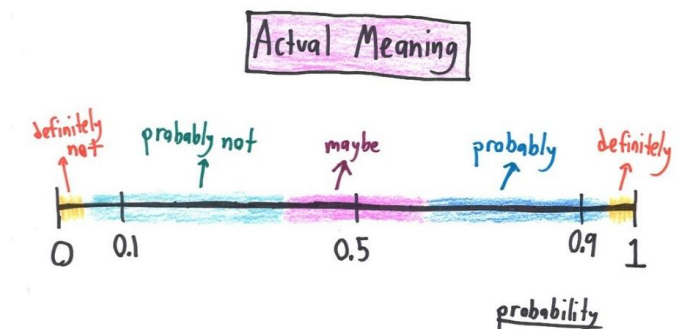


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 10 - Lecture No 18 – October 20, 2025



Announcements!

1. All lectures posted.
2. Scores for all 5 quizzes and MT posted.
3. **Quiz No 06** – Monday October 27, 2025 (DRV – Conditional probability and Total Probability)

Agenda for today

1. Unit 4 - Discrete Random Variables

Conditional Expectation

$$A = \{ Y = y \}$$

$$\mathbf{E}[X] = \sum_x x p_X(x) \quad \mathbf{E}[X | A] = \sum_x x p_{X|A}(x) \quad \mathbf{E}[X | Y = y] = \sum_x x p_{X|Y}(x | y)$$

Expected Value Rule

$$\mathbf{E}[g(X)] = \sum_x g(x) p_X(x) \quad \mathbf{E}[g(X) | A] = \sum_x g(x) p_{X|A}(x)$$

$$\mathbf{E}[g(X) | Y = y] = \sum_x g(x) p_{X|Y}(x | y)$$

Total probability and expectation theorems

$$Y = \{ y_1, y_2, \dots, y_n \} \quad A_i = \{ Y = y_i \}$$

- Partition of sample space in $A_1, A_2, \dots, A_n$
- $p_X(x) = P(A_1) p_{X|A_1}(x) + \dots + P(A_n) p_{X|A_n}(x)$

$$p_X(x) = \sum_y p_Y(y) p_{X|Y}(x|y)$$

- $\mathbf{E}[X] = P(A_1) \mathbf{E}[X | A_1] + \dots + P(A_n) \mathbf{E}[X | A_n]$

$$\mathbf{E}[X] = \sum_y p_Y(y) \mathbf{E}[X | Y = y]$$

- **Fine print:**

Also valid when Y is a discrete R.V. that ranges over an infinite set, as long as $\mathbf{E}[|X|] < \infty$

Independence

- Of two events: $\mathbf{P} (A \cap B) = \mathbf{P} (A) \mathbf{P} (B)$ $\mathbf{P} (A | B) = \mathbf{P} (A)$
- of a r.v. and an event: $\mathbf{P} (\underline{X = x \text{ and } A}) = \mathbf{P} (X = x) \mathbf{P} (A)$ for all x
 $\Rightarrow p_{X|A}(x) = p_X(x)$ for all x $\Rightarrow P (A | X = x) = P (A)$ for all x
- of two r.v.'s: $\mathbf{P} (X = x \text{ and } Y = y) = \mathbf{P} (X = x) \mathbf{P} (Y = y)$ for all x, y
 $\Rightarrow p_{X|Y}(x|y) = p_X(x)$
 $\Rightarrow p_{Y|X}(y|x) = p_Y(y)$
 $p_{X,Y}(x,y) = p_X(x) p_Y(y)$

X, Y, Z are independent if:

$$p_{X,Y,Z}(x,y,z) = p_X(x) p_Y(y) p_Z(z), \quad \text{for all } x, y, z$$

Example: independence and conditional independence

Y = Second roll

→

4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		
	1	2	3	4

X = First roll

↑

1/3	1/9	2/9
2/3	2/9	4/9
	1/3	2/3

- Independent? **No**

→ $P_X(1) = 3/20$

→ $P_{X|Y}(1|1) = 0$

- What if we condition on $X \leq 2$ and $Y \geq 3$?

→ $P_X(1) = 1/3$

→ $P_{X|Y}(1|3) = 1/3$

Yes



Exercise No 1

A Random Variable Z_n is defined as $Z_n = \min(X_1; X_2; \dots; X_n)$, where X_i 's are outcomes of identical and independently distributed fair six-sided die. As an example $Z_2 = \min(X_1; X_2)$ - meaning Z_2 is the minimum of the outcome of two independent throws of six-sided die X_1 and X_2 .

- (a) **(5 marks)** Calculate probability $P(1 < Z_{10} < 6)$.
- (b) **(5 marks)** Find the minimum value of n such that $P(Z_n = 1) \geq 0.6$



Exercise No 1

A Random Variable Z_n is defined as $Z_n = \min(X_1; X_2; \dots; X_n)$, where X_i 's are outcomes of identical and independently distributed fair six-sided die. As an example $Z_2 = \min(X_1; X_2)$ - meaning Z_2 is the minimum of the outcome of two independent throws of six-sided die X_1 and X_2 .

(a) (5 marks) Calculate probability $P(1 < Z_{10} < 6)$.

$$(a) \quad P(1 < Z_{10} < 6) = 1 - P(Z_{10} = 1) - P(Z_{10} = 6)$$

$$P(Z_{10} = 1) = 1 - P(Z_{10} \neq 1)$$

$$= 1 - P(Z_{10} \geq 2)$$

$$= 1 - P(\text{All 10 die are between 2-6})$$

$$P(Z_{10} = 1) = 1 - \left(\frac{5}{6}\right)^{10}$$

$$P(Z_{10} = 6) = P(\text{All 10 die are 6})$$

$$= \left(\frac{1}{6}\right)^{10}$$

$$P(Z_{10} = 6) = P(\text{All 10 die are 6})$$
$$= \left(\frac{1}{6}\right)^{10}$$

$$P(1 < Z_{10} < 6) = 1 - \left(1 - \left(\frac{5}{6}\right)^{10}\right) - \left(\frac{1}{6}\right)^{10}$$
$$= 1 - 1 + \left(\frac{5}{6}\right)^{10} - \left(\frac{1}{6}\right)^{10}$$
$$= \left(\frac{5}{6}\right)^{10} - \left(\frac{1}{6}\right)^{10} \quad \underline{\underline{0.1615}}$$



Exercise No 1

A Random Variable Z_n is defined as $Z_n = \min(X_1; X_2; \dots; X_n)$, where X_i 's are outcomes of identical and independently distributed fair six-sided die. As an example $Z_2 = \min(X_1; X_2)$ - meaning Z_2 is the minimum of the outcome of two independent throws of six-sided die X_1 and X_2 .

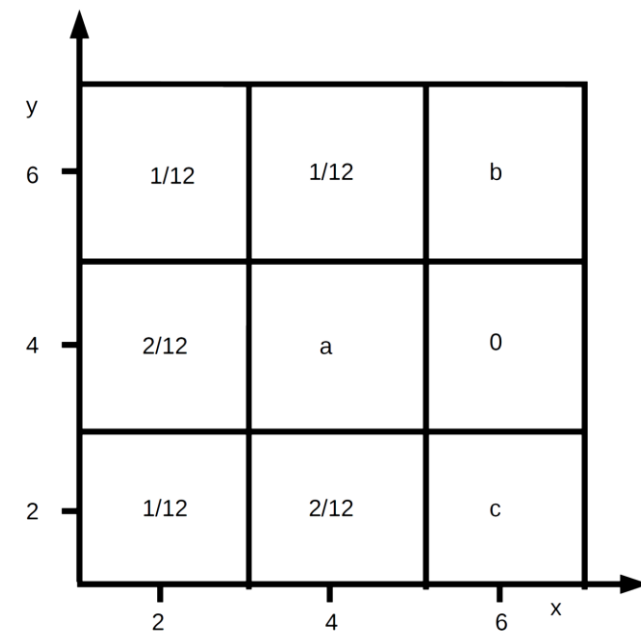
- (a) (5 marks) Calculate probability $P(1 < Z_{10} < 6)$.
- (b) (5 marks) Find the minimum value of n such that $P(Z_n = 1) \geq 0.6$

$$\begin{aligned} P(Z_n = 1) &= 1 - P(Z_n \neq 1) \\ &= 1 - P(\text{All } n \text{ throws are greater than } 1) \\ &= 1 - \left(\frac{5}{6}\right)^n \\ \Rightarrow 1 - \left(\frac{5}{6}\right)^n &\geq 0.6 \\ \left(\frac{5}{6}\right)^n &\leq 0.4 \Rightarrow n \log \frac{5}{6} \leq \log 0.4 \\ n &\geq \frac{-0.397}{-0.079} \\ n &\geq 5.02 \\ n &= 6 \end{aligned}$$

Exercise No 2

Sample Space for two events x and y with probabilities of event $x \cap y$ are given in Fig. 1

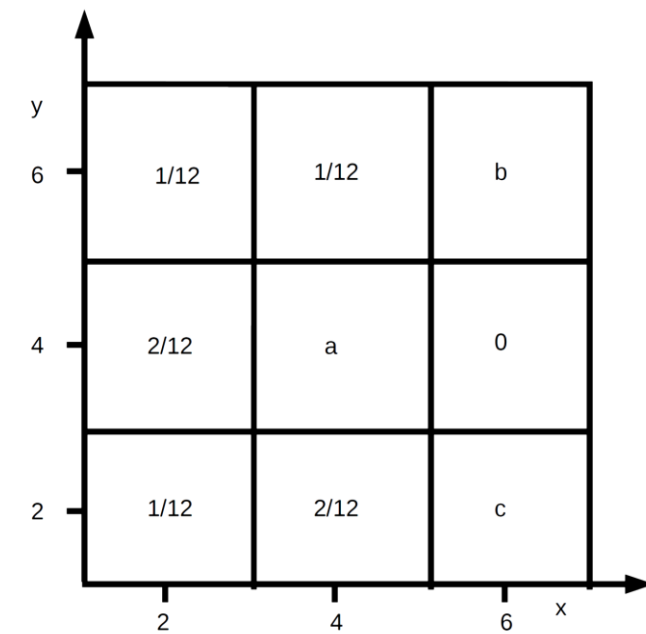
- (a) (**3 marks**) Calculate the probability that $x = 2$.
- (b) (**3 marks**) Find and sketch the probabilities $P_{y|x}(y|x = 2)$.
- (c) (**4 marks**) Find the probabilities (a, b, c) under which event x and y are independent each other. You need to provide proper justification and working for your answer.



Exercise No 2

Sample Space for two events x and y with probabilities of event $x \cap y$ are given in Fig. 1

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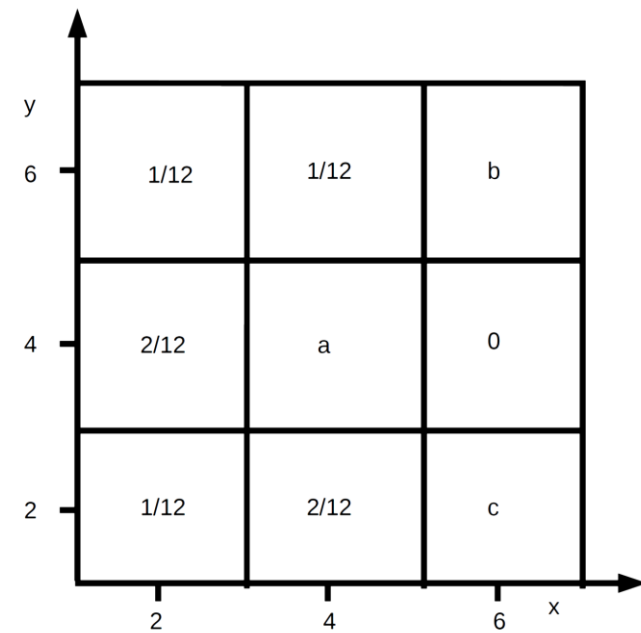


$$\begin{aligned} \text{(a)} \quad P(x=2) &= \sum_{y, x=2} P(x, y) \\ &= P(2, 2) + P(2, 4) + P(2, 6) \\ &= \frac{1}{12} + \frac{2}{12} + \frac{1}{12} = \frac{4}{12} \\ P(x=2) &= \frac{1}{3} \end{aligned}$$

Exercise No 2

Sample Space for two events x and y with probabilities of event $x \cap y$ are given in Fig. 1

- (a) **(3 marks)** Calculate the probability that $x = 2$.
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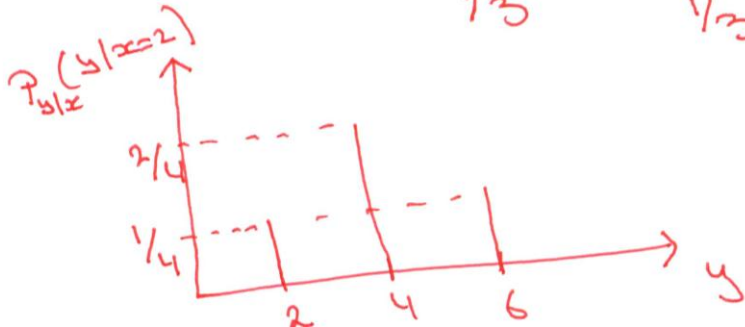


$$(b) \quad P(y|x=2) = \frac{P(y \cap (x=2))}{P(x=2) = 1/3}$$

$$y=2 \longrightarrow = \frac{P(2,2)}{1/3} = \frac{1/12}{1/3} = 1/4$$

$$y=4 \longrightarrow = \frac{P(2,4)}{1/3} = \frac{2/12}{1/3} = \frac{2}{4} = 1/2$$

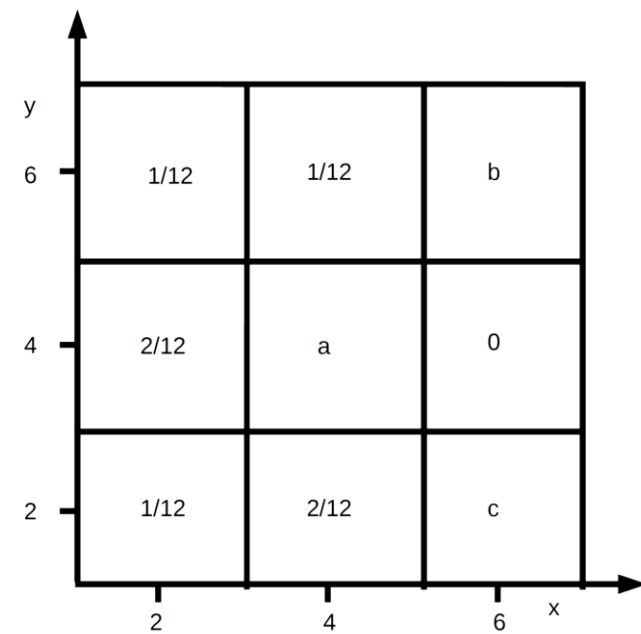
$$y=6 \longrightarrow = \frac{P(2,6)}{1/3} = \frac{1/12}{1/3} = 1/4$$



Exercise No 2

Sample Space for two events x and y with probabilities of event $x \cap y$ are given in Fig. 1

- (a) (3 marks) Calculate the probability that $x = 2$.
- (b) (3 marks) Find and sketch the probabilities $P_{y|x}(y|x = 2)$.
- (c) (4 marks) Find the probabilities (a, b, c) under which event x and y are independent each other. You need to provide proper justification and working for your answer.



c) When $x = 6$, y cannot be 4
 $\Rightarrow P(y=4|x=6) = 0$
however $P(y=4|x=2) = 2/4 \neq 0$
This implies that x, y can
never be independent and hence
there exists no value for
 a, b, c that makes x, y
Independent.

Independence and Expectations

- In general: $\mathbf{E}[g(X, Y)] \neq g(\mathbf{E}[X], \mathbf{E}[Y])$ *always true*
- Exceptions: $\mathbf{E}[aX + b] = a\mathbf{E}[X] + b$ $\mathbf{E}[X + Y + Z] = \mathbf{E}[X] + \mathbf{E}[Y] + \mathbf{E}[Z]$

If X, Y are **independent**:

$$\mathbf{E}[XY] = \mathbf{E}[X] \mathbf{E}[Y]$$

$g(X)$ and $h(Y)$ are also independent:

$$\mathbf{E}[g(X) h(Y)] = \mathbf{E}[g(X)] \mathbf{E}[h(Y)]$$

$$\mathbf{E}[g(X, Y)]$$

$$g(x, y) = xy$$

$$\rightarrow = \sum_x \sum_y (xy) p_{X,Y}(x,y) = \sum_x \sum_y (xy) p_X(x) p_Y(y)$$

$$\rightarrow = \sum_x x p_X(x) \sum_y y p_Y(y) = \mathbf{E}[X] \mathbf{E}[Y]$$

Question 1 - (15 points)

Let's be optimistic and assume that Pakistan cricket team has qualified for the T20 World Cup Semi Final. Pakistan's opponent in Semi Final will be West Indies, England, or Australia with a probability of 0.4, 0.4, and 0.2 respectively. The probability of Pakistan winning against West Indies is 0.6. The probability of Pakistan winning against England is 0.4. The probability of Pakistan winning against Australia is 0.6.

- a) (7.5 points) What is the probability that Pakistan will win the Semi-final.
- b) (7.5 points) Assume that you decided to completely switch off from cricket and did not follow the game. Later, you were told that Pakistan lost the Semi Final. What is the probability that Pakistan played the Semi Final game against England?

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Solution:

(a)

Let's assume the following naming conventions for various events,

$A_1 = \{ \text{Pakistan play West Indies in Semi Final} \}$

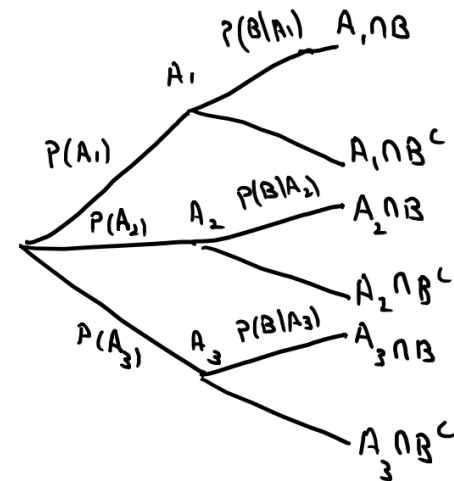
$A_2 = \{ \text{Pakistan play England in Semi Final} \}$

$A_3 = \{ \text{Pakistan play Australia in Semi Final} \}$

$B = \{ \text{Pakistan win the Semi Final} \}$

$B^c = \{ \text{Pakistan lose the Semi Final} \}$

$$\Rightarrow \begin{array}{lll} P(A_1) = 0.4 & P(A_2) = 0.4 & P(A_3) = 0.2 \\ P(B|A_1) = 0.6 & P(B|A_2) = 0.4 & P(B|A_3) = 0.6 \\ P(B^c|A_1) = 0.4 & P(B^c|A_2) = 0.6 & P(B^c|A_3) = 0.4 \end{array}$$



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By Total Probability Theorem:-

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B)$$

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$$

$$= (0.4)(0.6) + (0.4)(0.4) + (0.2)(0.6)$$

$$= 0.24 + 0.16 + 0.12$$

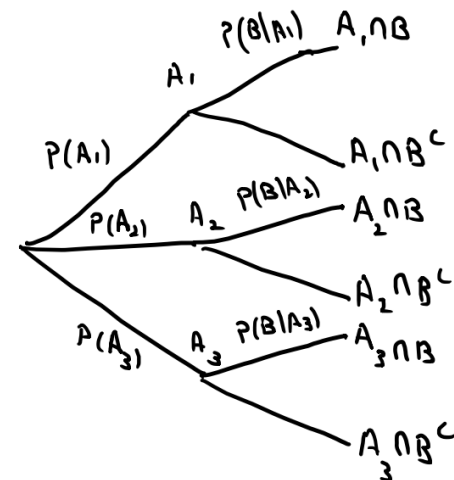
$$P(B) = 0.52 \quad \text{And}$$

$$P(B^c) = 1 - P(B) = 0.48$$

$$\Rightarrow P(A_1) = 0.4 \quad P(A_2) = 0.4 \quad P(A_3) = 0.2$$

$$P(B|A_1) = 0.6 \quad P(B|A_2) = 0.4 \quad P(B|A_3) = 0.6$$

$$P(B^c|A_1) = 0.4 \quad P(B^c|A_2) = 0.6 \quad P(B^c|A_3) = 0.4$$



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$$P(B^c) = 1 - P(B) = 0.48$$

According to Bayes' Rule,

$$P(A_2|B^c) = \frac{P(A_2)P(B^c|A_2)}{P(B^c)}$$

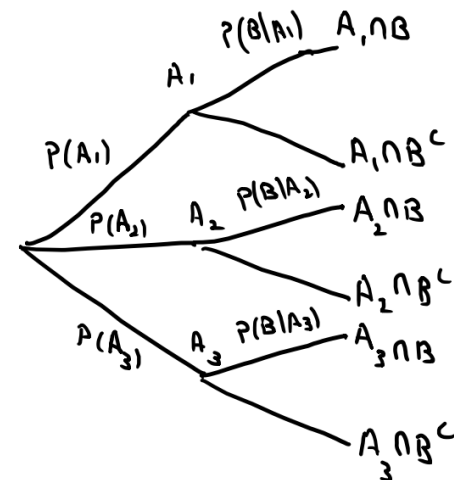
$$= \frac{(0.4)(0.6)}{0.48}$$

$$P(A_2|B^c) = 0.5 \quad \text{--- Ans (b)}$$

$$\Rightarrow P(A_1) = 0.4 \quad P(A_2) = 0.4 \quad P(A_3) = 0.2$$

$$P(B|A_1) = 0.6 \quad P(B|A_2) = 0.4 \quad P(B|A_3) = 0.6$$

$$P(B^c|A_1) = 0.4 \quad P(B^c|A_2) = 0.6 \quad P(B^c|A_3) = 0.4$$



Question 3 - (14 points)

Consider the following random variables X & Y:

X = Magnitude of the difference between the outcomes of two rolls of a fair 3-sided die.

(If the first roll results in 1 and second roll results in 3, the value taken by X is 2)

(If the first roll results in 3 and second roll results in 1, the value taken by X is 2)

Y = 2X

a) (4 points) Calculate and plot the PMF of X

b) (4 points) Calculate and plot the PMF of Y

c) (4 points) Calculate Mean of Y.

d) (4 points) Calculate Variance of Y

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- a) (4 points) Calculate and plot the PMF of X
- b) (4 points) Calculate and plot the PMF of Y
- c) (4 points) Calculate Mean of Y.
- d) (4 points) Calculate Variance of Y

Possible magnitudes of the difference between two rolls:

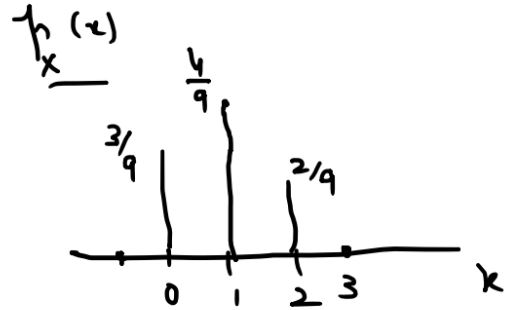
Roll 2 = 3	2	1	0
Roll 2 = 2	1	0	1
Roll 2 = 1	0	1	2
	Roll 1 = 1	Roll 1 = 2	Roll 1 = 3

Possible values that X can take = {0, 1, 2}

$P(X=0) = \frac{3}{9}$

$P(X=1) = \frac{4}{9}$

$P(X=2) = \frac{2}{9}$



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(d)

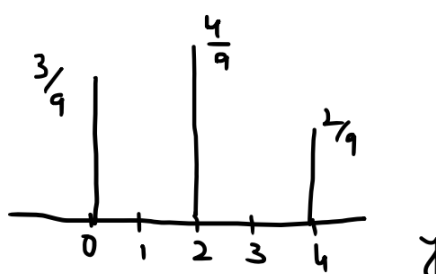
$$\text{var}(Y) = E[Y^2] - (\mu_Y)^2$$
$$E[Y^2] = \sum_y y^2 p_Y(y) \quad (\text{Expected Value Rule})$$
$$= 0^2 \left(\frac{3}{9}\right) + 2^2 \left(\frac{4}{9}\right) + 4^2 \left(\frac{2}{9}\right)$$
$$= \frac{16}{9} + \frac{32}{9}$$
$$E[Y^2] = \frac{48}{9}$$
$$\text{var}(Y) = \frac{48}{9} - \left(\frac{16}{9}\right)^2$$
$$\text{var}(Y) = 2.17$$

(b)

$$Y = 2X$$

Possible values of Y = {0, 2, 4}

$p_Y(y)$:



(c)

$$\mu_Y = E[Y] = \sum_y y p_Y(y)$$
$$= 0 \left(\frac{3}{9}\right) + 2 \left(\frac{4}{9}\right) + 4 \left(\frac{2}{9}\right)$$
$$\mu_Y = E[Y] = \frac{16}{9} = 0.17$$

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Consider the following random variables X & Y:

X = Magnitude of the difference between the outcomes of two rolls of a fair 3-sided die.

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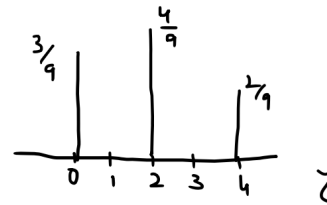
c) (4 points) Calculate Mean of Y.

d) (4 points) Calculate Variance of Y

$$Y = 2X$$

Possible values of Y = {0, 2, 4}

$p_Y(y)$:



$$\begin{aligned} \mu_Y = E[Y] &= \sum_y y p_Y(y) \\ &= 0\left(\frac{3}{9}\right) + 2\left(\frac{4}{9}\right) + 4\left(\frac{2}{9}\right) \end{aligned}$$

$$\mu_Y = E[Y] = \frac{16}{9} = 0.17$$

(d)

$$\text{var}(Y) = E[Y^2] - (\mu_Y)^2$$

$$\begin{aligned} E[Y^2] &= \sum_y y^2 p_Y(y) \quad (\text{Expected Value Rule}) \\ &= 0^2\left(\frac{3}{9}\right) + 2^2\left(\frac{4}{9}\right) + 4^2\left(\frac{2}{9}\right) \end{aligned}$$

$$= \frac{16}{9} + \frac{32}{9}$$

$$E[Y^2] = \frac{48}{9}$$

$$\text{var}(Y) = \frac{48}{9} - \left(\frac{16}{9}\right)^2$$

$$\text{var}(Y) = 2.17$$

Question 4 - (4 points)

Consider two random variables X & Y:

X = Temperature in Karachi on Oct 29, 2021 in Celsius ($^{\circ}\text{C}$)

Y = Temperature in Karachi on Oct 29, 2021 in Fahrenheit ($^{\circ}\text{F}$)

Assume that you have been provided the PMF of X and the corresponding PMF of Y.

Case 1: You take the mean of X and then convert this mean value to $^{\circ}\text{F}$

Case 2: You take the mean of Y

Will you get the same result in Case 1 and Case 2? Why or why not? Justify your answer.

(Hint: Celsius to Fahrenheit Conversion Formula: $^{\circ}\text{F} = 1.8^{\circ}\text{C} + 32$)

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Case 2: You take the mean of Y

Will you get the same result in Case 1 and Case 2? Why or why not? Justify your answer.

(Hint: Celsius to Fahrenheit Conversion Formula: $^{\circ}\text{F} = 1.8^{\circ}\text{C} + 32$)

If Y is a linear function of X , i.e.

$$Y = aX + b$$

$$E[Y] = aE[X] + b$$

Here,

$$Y = 1.8X + 32$$

$$\Rightarrow E[Y] = 1.8E[X] + 32$$

Case 2

Computation in Case 1

Therefore, you will get the same result in both cases.

Independence and Variances

- Always true: $\text{var}(aX + b) = a^2 \text{var}(X)$ $\text{var}(X + a) = \text{var}(X)$
- In general: $\text{var}(X + Y) \neq \text{var}(X) + \text{var}(Y)$

If X, Y are **independent**: $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$

assume

$$E[X] = 0 ; E[Y] = 0$$

$$E[X]E[Y] = 0$$

$$\begin{aligned} \Rightarrow \text{var}(X + Y) &= E[(X + Y)^2] = E[X^2 + 2XY + Y^2] \\ \Rightarrow &= E[X^2] + 2E[XY] + E[Y^2] = E[X^2] + E[Y^2] = \text{var}(X) + \text{var}(Y) \end{aligned}$$

- Examples:

➤ If $X = Y$: $\text{var}(X + Y) = \text{var}(2X) = 4\text{var}(X)$

➤ If $X = -Y$: $\text{var}(X + Y) = \text{var}(0) = 0$

➤ If X, Y independent: $\text{var}(X - 3Y) = \text{var}(X) + \text{var}(-3Y) = \text{var}(X) + 9\text{var}(Y)$

Variance of the binomial

- X : binomial with parameters n, p
 - ❖ number of successes in n independent trials

$X_i = 1$ if i^{th} trial is a success; p
(indicator variable)
 $X_i = 0$ otherwise $1-p$

independent

$$X = X_1 + \dots + X_n$$

$$\rightarrow \text{var}(X) = \text{var}(X_1) + \text{var}(X_2) + \dots + \text{var}(X_n)$$

$$\rightarrow \text{var}(X) = n \text{var}(X_1) = np(1-p)$$

The variance of the geometric PMF

X: geometric parameter with p

- Given: $X > 1$, $X - 1$ has the same geometric PMF (unconditional PMF of X)
- Given $X > 1$, conditional PMF of X: same as unconditional PMF of $X + 1$

$$\rightarrow E[X \mid X > 1] = 1 + E[X] \quad E[X^2 \mid X > 1] = E[(X + 1)^2]$$

$$\begin{aligned} \rightarrow E[X^2] &= P(X=1) E[X^2 \mid X=1] + P(X > 1) E[X^2 \mid X > 1] \\ &= p \cdot 1 + (1-p) (E[X^2] + 2E[X] + 1) \end{aligned}$$

$\xrightarrow{\hspace{10em}} 1/p$

$$\text{Algebra: } E[X^2] = \frac{2}{p^2} - \frac{1}{p}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

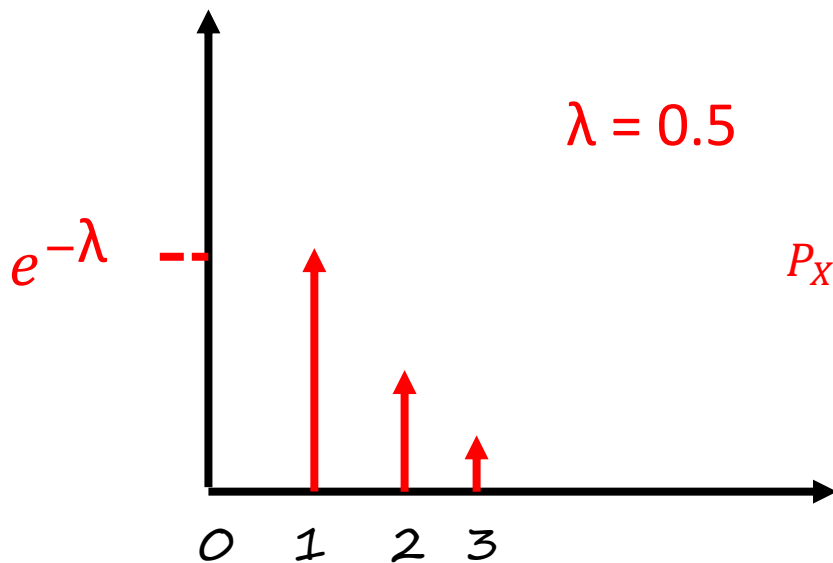
$$\text{Var}(X) = \frac{1-p}{p^2}$$

Poisson RV

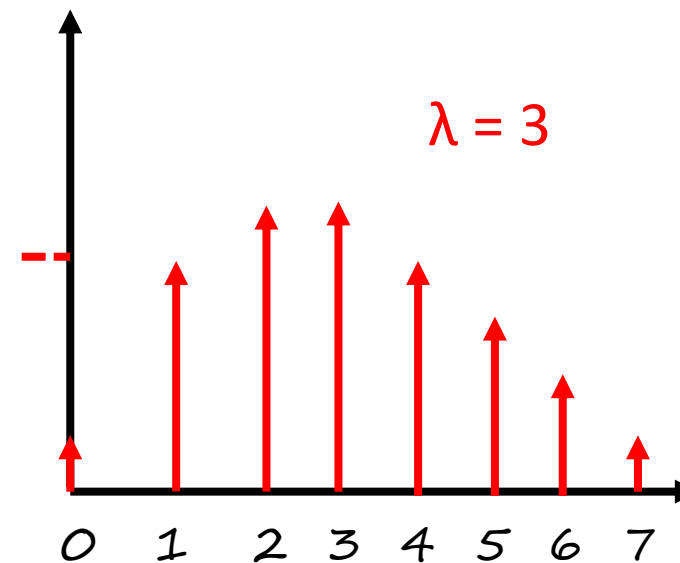
Parameter: λ (A positive characterizing the PMF)

$$P_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}, K = 0, 1, 2, 3$$

Note: $\sum_{K=0}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} = e^{-\lambda} (1 + \lambda + \frac{\lambda^2}{2} + \dots) = e^{-\lambda} * e^{\lambda} = 1$



Note: When $\lambda < 1$, PMF, is monotonically decreasing with K



Note: When $\lambda > 1$, PMF first increases and then decreases

Models: Probability of occurrence of events in a fixed interval of time or space if these events occur with a known constant rate and independent of time of last event.

Poisson RV: Example

$$P_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

An IC manufacturing company provides 100 IC's per day. Defective IC's are at a fixed rate of 3 IC's/day. Probability of R defective IC's in a given day.

$$P_X(0) = e^{-3} \approx 0.05$$

$$P_X(3) = e^{-3} \frac{3^3}{3!} = 0.05 * 4.5 = 0.225$$

How do you find $P(X > 3)$?

Useful Modeling

- a) No. of phone calls received by a call center (per hour, per day, or per anytime unit)
- b) No. of photons hitting a detector in a particular time interval